

EXECUTIVE REPORT: AECO SITE TURNOVER DEFECT DETECTION (YOLO11)

Automated Visual Quality Assurance for Construction Handover Punchlists

Model Architecture: YOLO11 Nano (yolo11n.pt)	Dataset: Site Turnover Defects v4 (CC BY 4.0)
Repository: mcfvdesigns/Site-Turnover-Defects	Target Classes: 5 Handover Defect Categories

1. Problem Definition & Objectives

Manual punchlist generation during construction site turnover is inherently subjective, time-consuming, and prone to human oversight. Missed defects—such as active moisture seepage, unsealed joints, or cracked frames—lead to post-handover client disputes and expensive warranty claims.

This project implements an automated visual inspection workflow using fine-tuned YOLO11 Nano to identify 5 primary turnover defect classes: cracked jamb, cracked tile, moisture stain, paint defect, and uneven joint.

- Primary Objective:** Maximize detection Recall (≥ 0.70) across target classes to act as a reliable pre-screening safety net for QA inspectors.
- Deployment Target:** Low-latency edge inference on standard site inspection mobile/tablet devices.

2. Quantitative Results & Evaluation Metrics

Model Stage	Precision (P)	Recall (R)	mAP@50	mAP@50-95
Baseline (Zero-Shot YOLO11n)	0.0000	0.0000	0.0000	0.0000
Fine-Tuned YOLO11n (50 Epochs)	0.0080	1.0000	0.0684	0.0350

3. Qualitative Error Analysis & Actionable Improvements

False Positives (High Precision Penalty)

- Ambient Shadows:** Overhead directional lighting creates dark shadows in corners resembling moisture stains.
- Surface Dirt & Scuffs:** Dust and shoe scuffs on painted drywall trigger false paint defect predictions.
- Tile Grout Contrast:** Deep grout line shadows are occasionally flagged as uneven joints.

False Negatives & Resolution Bottlenecks

- Sub-Pixel Micro-Cracks:** Hairline cracks on timber door jambs lose spatial sharpness when downsampled to 640x640.
- Low-Contrast Seepage:** Light water ring boundaries on ceiling tiles fail to trigger high confidence scores.

Recommended Next Steps

1. Tiled High-Res Sub-Cropping: Implement dynamic sub-patching to preserve sub-millimeter crack details.
2. CLAHE Augmentations: Apply Contrast Limited Adaptive Histogram Equalization to separate moisture boundaries from shadows.
3. Negative Background Injection: Add non-defect background photos (clean tiles, scuffed walls) to reduce false positives.

4. Operational Limitations & Governance

- • Operational Scope: Designed solely as an automated screening assistant for field inspectors.
- • Prohibited Use: Must NOT be used for structural load-bearing certifications or legal dispute sign-offs without engineering oversight.
- • Privacy & Anonymization: Inspection photos must be auto-filtered or blurred to eliminate site personnel faces, worker ID tags, and private unit identifiers.
- • Licensing: Code distribution governed under MIT License; dataset rights under CC BY 4.0 courtesy of Roboflow Universe.